PK-Sim® Ontogeny Database

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1 Aim

This document represents a summary of literature data to ontogeny of enzymes, and other ADME relevant proteins as well as fit results used for implementation of ontogeny functions in PK-Sim®.

2 Data management

2.1 Literature data

Used literature data are referenced in the respective enzyme sections.

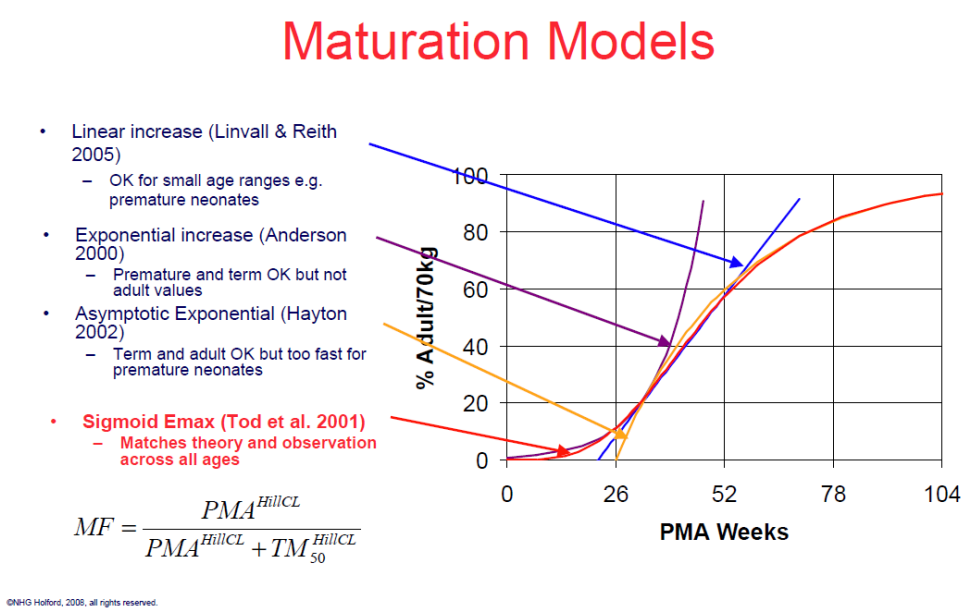
2.2 Fit results

Fit results and figures are illustrated in the respective enzyme sections.

3 Methods

A maturation model based on a sigmoidal Emax model (corresponding to the Hill-equation), published by Tod et al. 2008 (Clin Pharmacokinet. 47(4):231-43) presents the best description of age dependent development of e.g. proteins (Fig. 1). This model was also presented at the EMEA Workshop on Modelling in Paediatric Medicines in 2008 (Presentation 'Prof. N. Holford, 2008, 'Mechanism-Based Concepts of Size and Maturity', www.ema.europa.eu/docs/en_GB/document_library/Agenda/2009/11/WC500010017.pdf).

Figure 1. Slide taken from 'http://www.ema.europa.eu/docs/en_GB/document_library/Presentation/2009/11/WC500009792.pdf')



Fitting of ontogeny data was therefore performed using the following Hill equation (Equation 1):

$$A = PMA^n/(PMA^n + A_{0.5}^n) \quad (1)$$

With:
PMA = Post-menstrual age in weeks
A = Relative activity at PMA

$A_{0.5}$ = PMA (in weeks) at 50 % activity compared to adult
n = Hill coefficient

In the case of proteins showing decreased expression with increasing age, the following inverse fit function was applied (Equation 2):

$$A = 1 - PMA^n / (A_{0.5}^{\ n} + PMA^n) + offset \quad (2)$$

With offset = activity in adults

For fitting the ontogeny including variability, constituting variability in ontogeny as well as activity, a virtual population with 10000 individuals was created. Fitted were then the geometric mean and geometric standard deviation of the Hill coefficient (GeoSD n), the geometric mean and geometric standard deviation of PMA at 50% activity compared to adult (GeoSD $A_{0.5}$), and a geometric standard deviation as an activity variability factor (GeoSD adult).

All illustrated figures are shown in full age-length and a zoom-in version in the first period after birth. The presented tables are the implemented tables in PK-Sim, which are created from the fitted Hill equations.

For the ontogenies of CES1 and P-gp, no fitting was performed but ontogeny functions from the literature were selected and modified as described in the respective sections.

4 Results

4.3 CYP Enzymes in the liver

A.1 CYP1A2 in liver

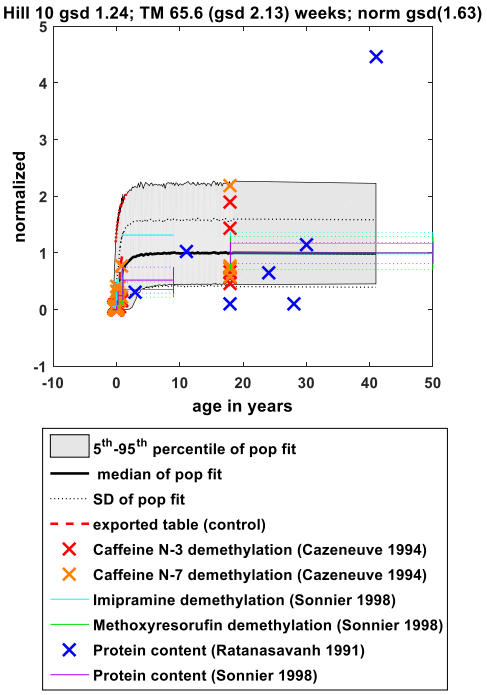
A.1.1 Original data

Reference	Used for fit
Br J Clin Pharmacol. 1994 May;37(5):405-12.; Biotransformation of caffeine in human liver microsomes from foetuses, neonates, infants and adults.;Cazeneuve C, Pons G, Rey E, Treluyer JM, Cresteil T, Thiroux G, D'Athis P, Olive G. Digital Object Identifyer (DOI): 10.1111/j.1365-2125.1994.tb05706.x → Table 2 (Caffeine N-3 demethylation, and Caffeine N-7 demethylation)	Yes
Eur J Biochem. 1998 Feb 1;251(3):893-8.; Delayed ontogenesis of CYP1A2 in the human liver.; Sonnier M, Cresteil T. Digital Object Identifyer (DOI): 10.1046/j.1432-1327.1998.2510893.x → Figure 2a → Figure 2b → Figure 3	 Yes Yes Yes
Hepatology. 1991 Jun;13(6):1142-51.; Intralobular distribution and quantitation of cytochrome P-450 enzymes in human liver as a function of age.; Ratanasavanh D1, Beaune P, Morel F, Flinois JP, Guengerich FP, Guillouzo A. Digital Object Identifyer (DOI): 10.1002/hep.1840130622 →Table 2 (protein content)	 Yes

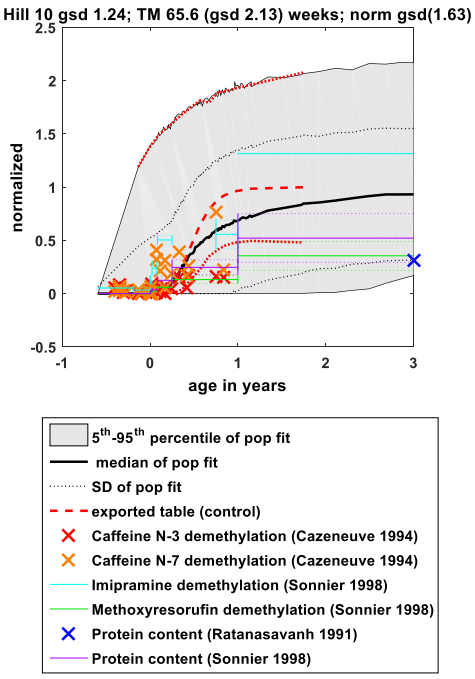
A.1.2 Fit results

A.1.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.1.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

Fit Result:
n = 10.015
$A_{0.5}$ = 65.639

GeoSD n = 1.241
GeoSD $A_{0.5}$ = 2.130
GeoSD adult = 1.628

A.1.2.3 Description for PK-Sim®

Values for CYP1A2 ontogeny are based on information to age dependency of Caffeine N-3 demethylation, Caffeine N-7 demethylation, Methoxyresorufin demethylation, Imipramine demethylation, and protein content, derived from the papers as mentioned in table A.1.1.

A.1.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP1A2	Standard Deviation
0.629999995	0.00099	74.55000305
0.769999981	0.007	24.93000031
0.879999995	0.026000001	11.80000019
0.980000019	0.071000002	6.539999962
1.049999952	0.140000001	4.420000076
1.139999986	0.25999999	3.150000095
1.340000033	0.649999976	1.870000005
1.440000057	0.790000021	1.639999986
1.549999952	0.879999995	1.590000033
1.610000014	0.920000017	1.549999952
1.74000001	0.959999979	1.529999971
1.929999948	0.99000001	1.519999981
2.519999981	1	1.559999943

A.2 CYP2C18 in liver

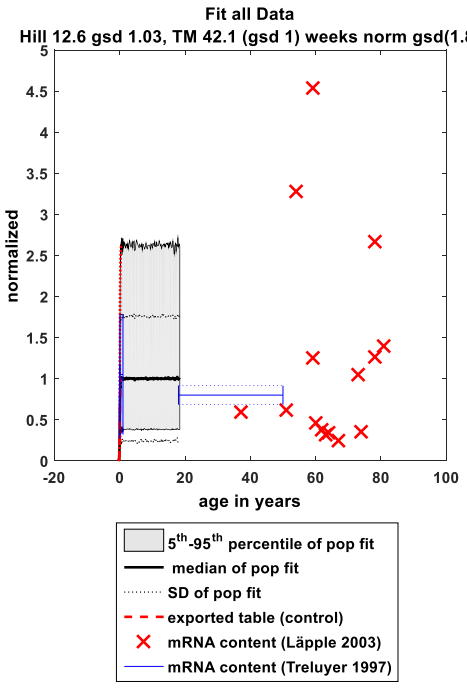
A.2.1 Original data

Reference	Used for fit
Pharmacogenetics. 1997 Dec;7(6):441-52.; Developmental expression of CYP2C and CYP2C-dependent activities in the human liver: in-vivo/in-vitro correlation and inducibility.; Treluyer JM, Gueret G, Cheron G, Sonnier M, Cresteil T. Pubmed ID (PMID): 9429229 (DOI not available) → Figure 5 (mRNA expression)	Yes
Pharmacogenetics. 2003 Sep;13(9):565-75.; Differential expression and function of CYP2C isoforms in human intestine and liver.; Läpple F, von Richter O, Fromm MF, Richter T, Thon KP, Wisser H, Griese EU, Eichelbaum M, Kivistö KT. Digital Object Identifyer (DOI): 10.1097/01.fpc.0000054122.14659.1e Pubmed ID (PMID): 12972955 (DOI not functioning) →Table 3 (mRNA expression)	Yes

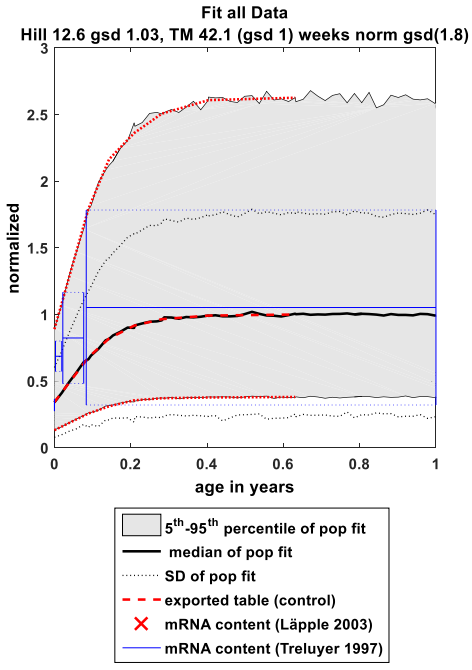
A.2.2 Fit results

A.2.2.1 Figure

Full range-plot



Zoom-in, x-axis



Comment: For the intestine, the geometric standard deviation in activity observed in adults ('GeoSD adult') has been separately fitted and implemented with a GeoSD of 2.177.

A.2.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

Fit Result:
n = 12.590
$A_{0.5}$ = 42.134

GeoSD n = 1.034
GeoSD $A_{0.5}$ = 1.002
GeoSD adult = 1.795

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.2.2.3 Description for PK-Sim®

Values for CYP2C18 ontogeny are based on information to age dependency of mRNA expression, derived from the papers as mentioned in table A.2.1.

A.2.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP2C18	Standard Deviation
0.469999999	0.00098	1.870000005
0.600000024	0.022	1.809999943
0.680000007	0.090999998	1.789999962
0.730000019	0.200000003	1.799999952
0.769999981	0.340000004	1.789999962
0.860000014	0.680000007	1.809999943
0.910000026	0.819999993	1.799999952
0.980000019	0.920000017	1.779999971
1.049999952	0.959999979	1.789999962
1.169999957	0.99000001	1.799999952
1.399999976	1	1.799999952

A.3 CYP2C19 in liver

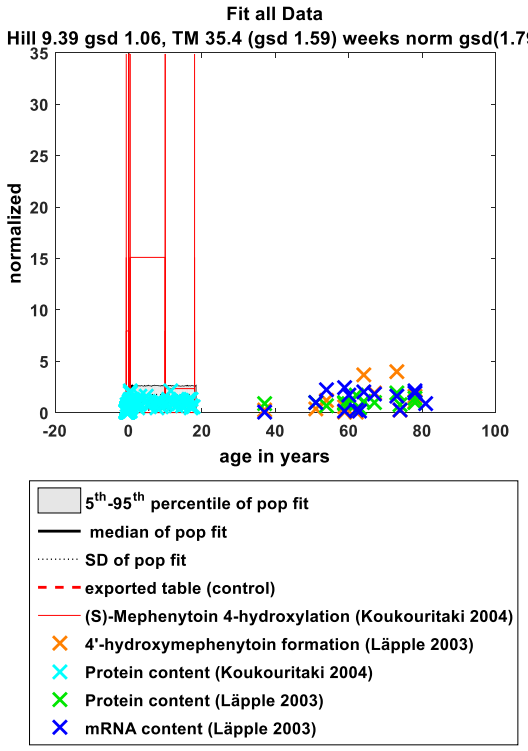
A.3.1 Original data

Reference	Used for fit
J Pharmacol Exp Ther. 2004 Mar;308(3):965-74. Epub 2003 Nov 21.; Developmental expression of human hepatic CYP2C9 and CYP2C19.; Koukouritaki SB, Manro JR, Marsh SA, Stevens JC, Rettie AE, McCarver DG, Hines RN. Digital Object Identifyer (DOI): 10.1124/jpet.103.060137 → Figure 3 (protein content) → Table 2 ((S)-Mephenytoin 4-hydroxylation)	 Yes Yes
Pharmacogenetics. 2003 Sep;13(9):565-75.; Differential expression and function of CYP2C isoforms in human intestine and liver.; Läpple F, von Richter O, Fromm MF, Richter T, Thon KP, Wisser H, Griese EU, Eichelbaum M, Kivistö KT. Digital Object Identifyer (DOI): 10.1097/01.fpc.0000054122.14659.1e Pubmed ID (PMID): 12972955 (DOI not functioning) → Table 3 (mRNA expression) → Table 6 (protein content)	 Yes Yes

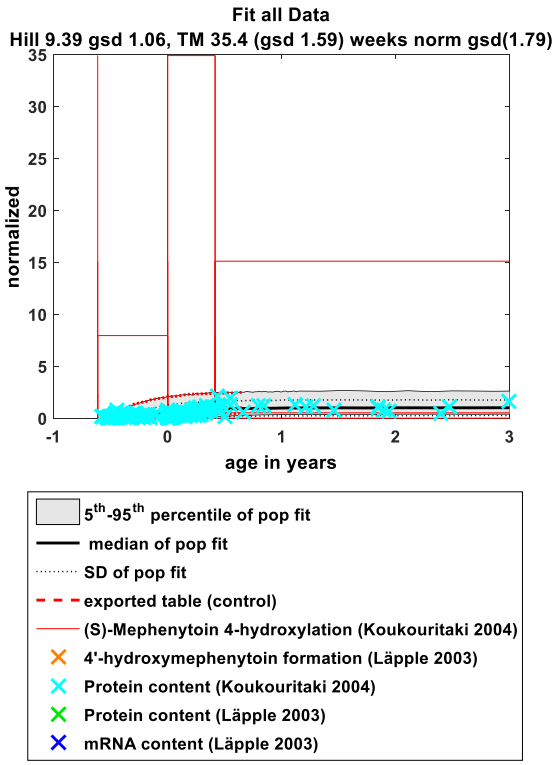
A.3.2 Fit results

A.3.2.1 Figure

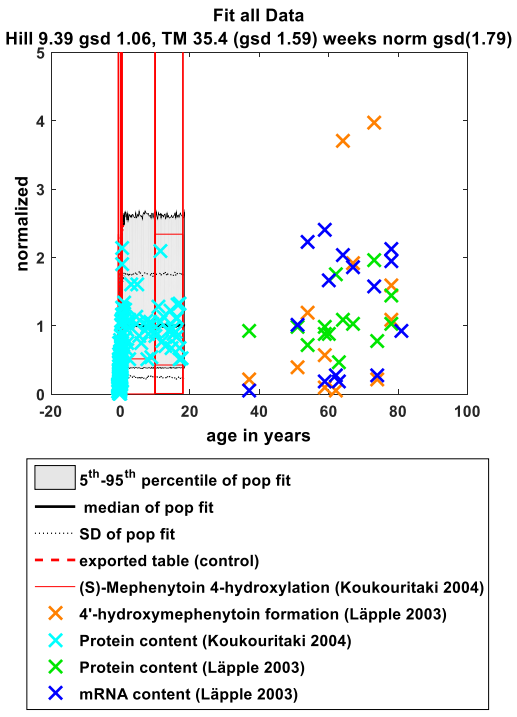
Full range-plot



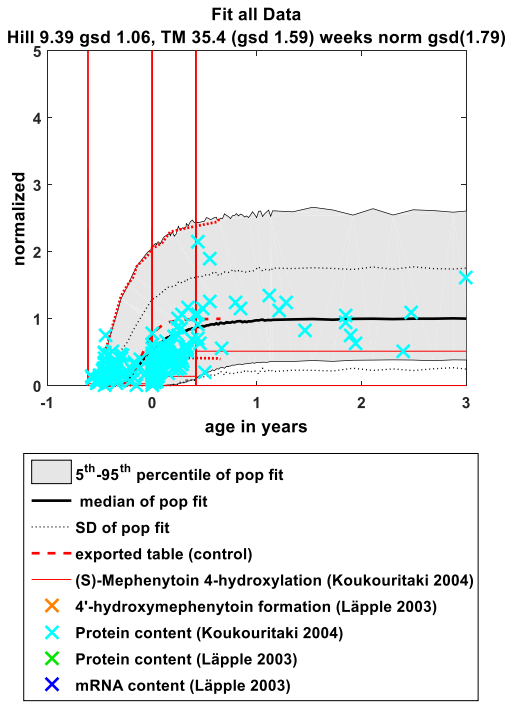
Zoom-in, x-axis



Zoom-in, y-axis



Zoom-in, x-axis and y-axis



Comment:

For the intestine, the geometric standard deviation in activity observed in adults ('GeoSD adult') has been separately fitted and implemented with a GeoSD of 2.883.

A.3.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

Fit Result:
n = 9.390
$A_{0.5}$ = 35.447

GeoSD n = 1.055
GeoSD $A_{0.5}$ = 1.592
GeoSD adult = 1.793

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.3.2.3 Description for PK-Sim®

Values for CYP2C1* ontogeny are based on information to age dependency of mRNA expression, protein content and (S)-Mephenytoin 4-hydroxylation derived from the papers as mentioned in table A.3.1.

A.3.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP2C19	Standard Deviation
0.33	0.001	46.35
0.47	0.03	10.04
0.53	0.09	5.56
0.58	0.18	3.76
0.62	0.3	2.96
0.72	0.64	1.96
0.77	0.76	1.84
0.81	0.83	1.75
0.87	0.91	1.68
0.94	0.95	1.71
1.09	0.99	1.69
1.39	1	1.72
1.42	1	1.75

A.4 CYP2C8 in liver

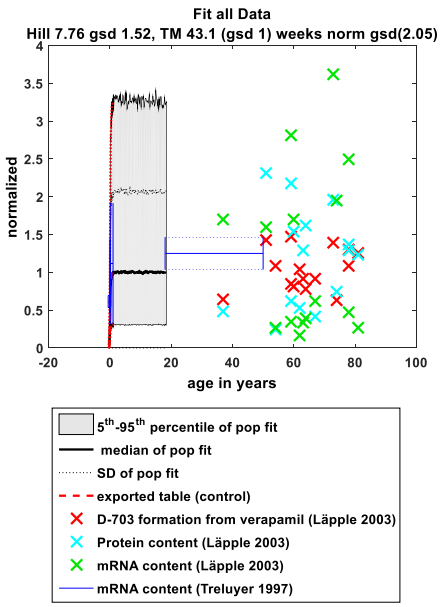
A.4.1 Original data

Reference	Used for fit
Pharmacogenetics. 1997 Dec;7(6):441-52.; Developmental expression of CYP2C and CYP2C-dependent activities in the human liver: in-vivo/in-vitro correlation and inducibility.; Treluyer JM, Gueret G, Cheron G, Sonnier M, Cresteil T. Pubmed ID (PMID): 9429229 (DOI not available) → Figure 5 (mRNA expression)	 Yes
Pharmacogenetics. 2003 Sep;13(9):565-75.; Differential expression and function of CYP2C isoforms in human intestine and liver.; Läpple F, von Richter O, Fromm MF, Richter T, Thon KP, Wisser H, Griese EU, Eichelbaum M, Kivistö KT. Digital Object Identifyer (DOI): 10.1097/01.fpc.0000054122.14659.1e Pubmed ID (PMID): 12972955 (DOI not functioning) →Table 3 (mRNA expression) →Table 4(protein content, D-703 formation from verapamil)	 Yes Yes

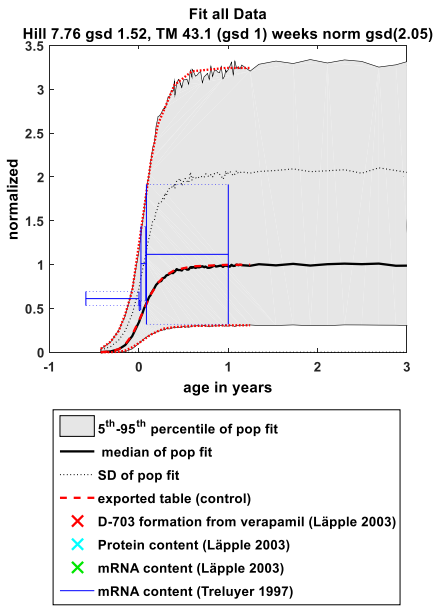
A.4.2 Fit results

A.4.2.1 Figure

Full range-plot



Zoom-in, x-axis



Comment:

For the intestine, the geometric standard deviation in activity observed in adults ('GeoSD adult') has been separately fitted and implemented with a GeoSD of 2.903

A.4.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

Fit Result:
n = 7.758
$A_{0.5}$ = 43.084

GeoSD n = 1.522
GeoSD $A_{0.5}$ = 1.000
GeoSD adult = 2.054

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.4.2.3 Description for PK-Sim®

Values for CYP2C8 ontogeny are based on information to age dependency of mRNA expression, protein content and D-703 formation from verapamil, derived from the papers as mentioned in table A.4.1.

A.4.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP2C8	Standard Deviation
0.34	0.001	9.23
0.45	0.0087	4.92
0.54	0.035	3.23
0.61	0.089	2.62
0.69	0.2	2.15
0.77	0.36	2.12
0.9	0.66	2.06
0.98	0.79	2.06
1.05	0.87	2.03
1.17	0.93	2.05
1.36	0.98	2.06
1.74	1	2.05

A.5 CYP2C9 in liver

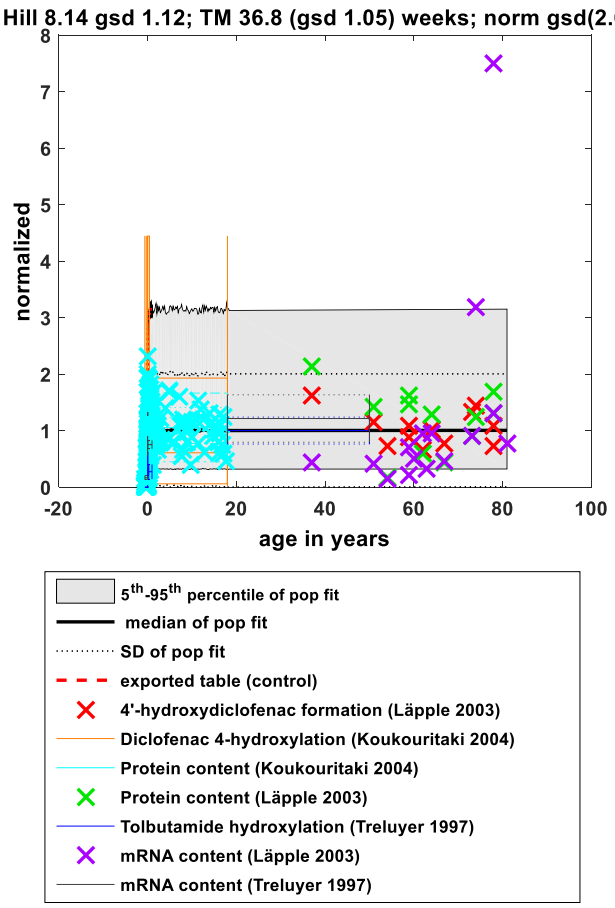
A.5.1 Original data

Reference	Used for fit
J Pharmacol Exp Ther. 2004 Mar;308(3):965-74. Epub 2003 Nov 21.; Developmental expression of human hepatic CYP2C9 and CYP2C19.; Koukouritaki SB, Manro JR, Marsh SA, Stevens JC, Rettie AE, McCarver DG, Hines RN. Digital Object Identifyer (DOI): 10.1124/jpet.103.060137 → Figure 2 (protein content) → Table 1 (Diclofenac 4-hydroxylation) → Figure 5	 Yes Yes No
Pharmacogenetics. 1997 Dec;7(6):441-52.; Developmental expression of CYP2C and CYP2C-dependent activities in the human liver: in-vivo/in-vitro correlation and inducibility.; Treluyer JM, Gueret G, Cheron G, Sonnier M, Cresteil T. Pubmed ID (PMID): 9429229 (DOI not available) → Figure 3 (Tolbutamide hydroxylation) → Figure 5 (mRNA expression)	 Yes Yes
Pharmacogenetics. 2003 Sep;13(9):565-75.; Differential expression and function of CYP2C isoforms in human intestine and liver.; Läpple F, von Richter O, Fromm MF, Richter T, Thon KP, Wisser H, Griese EU, Eichelbaum M, Kivistö KT. Digital Object Identifyer (DOI): 10.1097/01.fpc.0000054122.14659.1e Pubmed ID (PMID): 12972955 (DOI not functioning) →Table 3 (mRNA expression) →Table 5 (4'hydroxydiclofenac formation, protein content)	 Yes Yes

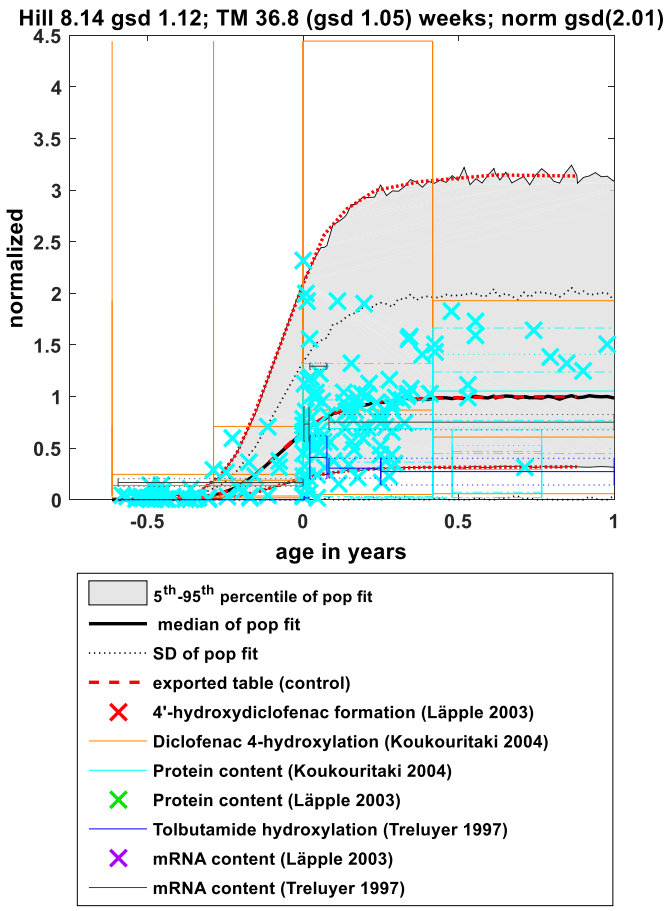
A.5.2 Fit results

A.5.2.1 Figure

Full range-plot



Zoom-in, x-axis



Comment:

For the intestine, the geometric standard deviation in activity observed in adults (‘GeoSD adult’) has been separately fitted and implemented with a GeoSD of 4.016

A.5.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

Fit Result:
n = 8.135
$A_{0.5}$ = 36.773

GeoSD n = 1.118
GeoSD $A_{0.5}$ = 1.049
GeoSD adult = 2.006

A.5.2.3 Description for PK-Sim®

Values for CYP2C9 ontogeny are based on information to age dependency of mRNA expression, protein content, tolbutamide hydroxylation, and 4'hydroxydiclofenac formation, derived from the papers as mentioned in table A.5.1.

A.5.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP2C9	Standard Deviation
0.3	0.00098	2.91
0.44	0.019	2.4
0.51	0.061	2.33
0.55	0.12	2.22
0.62	0.24	2.15
0.77	0.66	2.01
0.84	0.8	2.03
0.91	0.89	2.02
1	0.94	2.02
1.15	0.98	2.01
1.39	1	2.01

A.6 CYP2D6 in liver

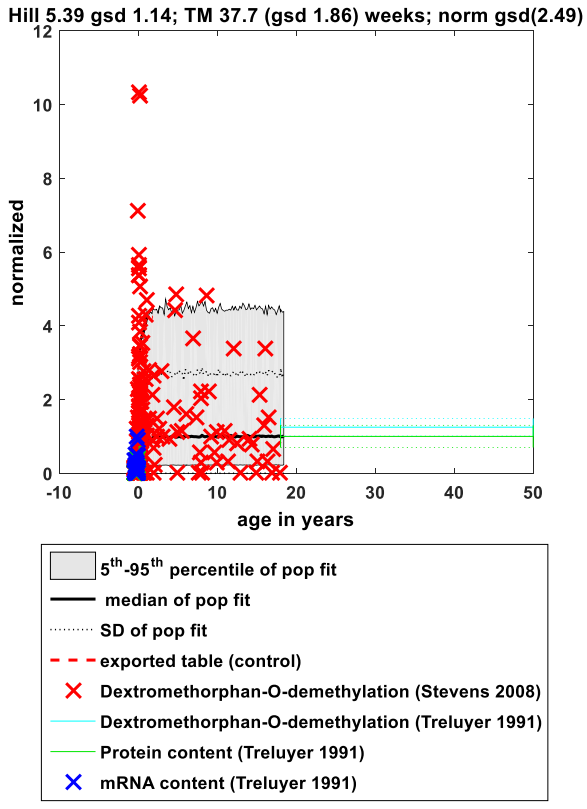
A.6.1 Original data

Reference	Used for fit
Drug Metab Dispos. 2008 Aug;36(8):1587-93. doi: 10.1124/dmd.108.021873. Epub 2008 May 12.; Developmental changes in human liver CYP2D6 expression.; Stevens JC, Marsh SA, Zaya MJ, Regina KJ, Divakaran K, Le M, Hines RN. Digital Object Identifyer (DOI): 10.1124/dmd.108.021873 → Figure 1A (Dextromethorphan-O-demethylation) → Figure 1B (Dextromethorphan-O-demethylation) → Figure 1C (Dextromethorphan-O-demethylation)	 Yes Yes Yes
Eur J Biochem. 1991 Dec 5;202(2):583-8.; Expression of CYP2D6 in developing human liver.; Treluyer JM, Jacqz-Aigrain E, Alvarez F, Cresteil T. Digital Object Identifyer (DOI): 10.1111/j.1432-1033.1991.tb16411.x → Figure 1 (protein content) → Figure 5 (Dextromethorphan-O-demethylation)	 Yes Yes

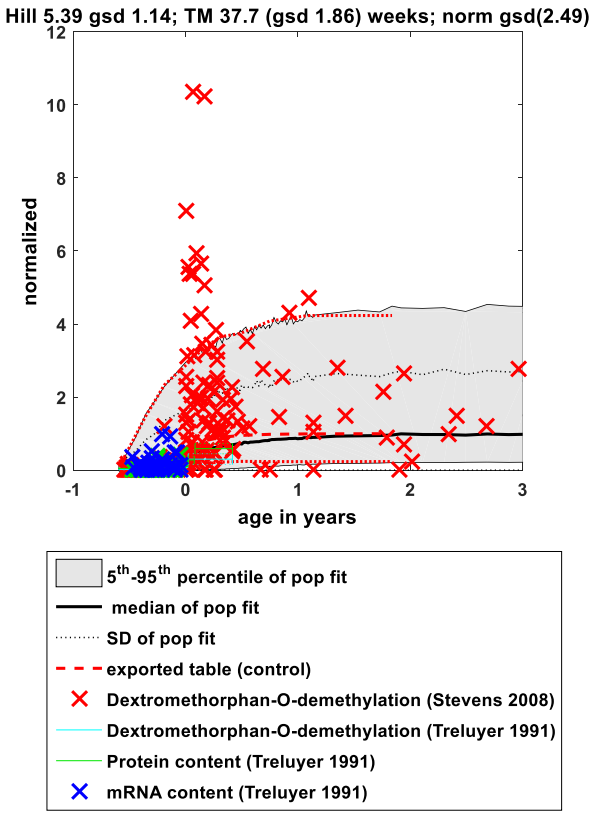
A.6.2 Fit results

A.6.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.6.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

Fit Result:
n = 5.391
$A_{0.5}$ = 37.706

GeoSD n = 1.139
GeoSD $A_{0.5}$ = 1.859
GeoSD adult = 2.488

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.6.2.3 Description for PK-Sim®

Values for CYP2D6 ontogeny are based on information to age dependency of protein content and dextromethorphan-O-demethylation activity, derived from the papers as mentioned in table A.6.1.

A.6.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP2D6	Standard Deviation
0.2	0.001	27.88
0.32	0.012	14.29
0.41	0.046	8.25
0.49	0.1	5.76
0.58	0.23	4.13
0.77	0.58	2.65
0.86	0.71	2.46
0.96	0.82	2.38
1.09	0.9	2.35
1.26	0.95	2.3
1.52	0.98	2.38
1.86	0.99	2.41
2.61	1	2.41

A.7 CYP2E1 in liver

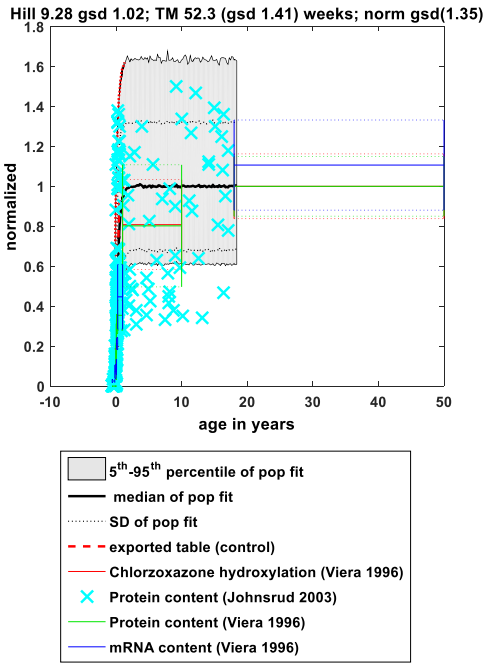
A.7.1 Original data

Reference	Used for fit
J Pharmacol Exp Ther. 2003 Oct;307(1):402-7.; Human hepatic CYP2E1 expression during development.; Johnsrud EK, Koukouritaki SB, Divakaran K, Brunengraber LL, Hines RN, McCarver DG. Digital Object Identifyer (DOI): 10.1124/jpet.102.053124 Pubmed ID (PMID): 12972955 (DOI not functioning) → Figure 2A (protein content) → Figure 2B (protein content) → Figure 3 (protein content)	 Yes Yes Yes
Eur J Biochem. 1996 Jun 1;238(2):476-83.; Developmental expression of CYP2E1 in the human liver. Hypermethylation control of gene expression during the neonatal period.; Vieira I, Sonnier M, Cresteil T. Digital Object Identifyer (DOI): 10.1111/j.1432-1033.1996.0476z.x → Figure 1 (protein content) → Figure 2 (Chlorzoxazone hydroxylation) → Figure 3 (mRNA expression)	 Yes Yes Yes

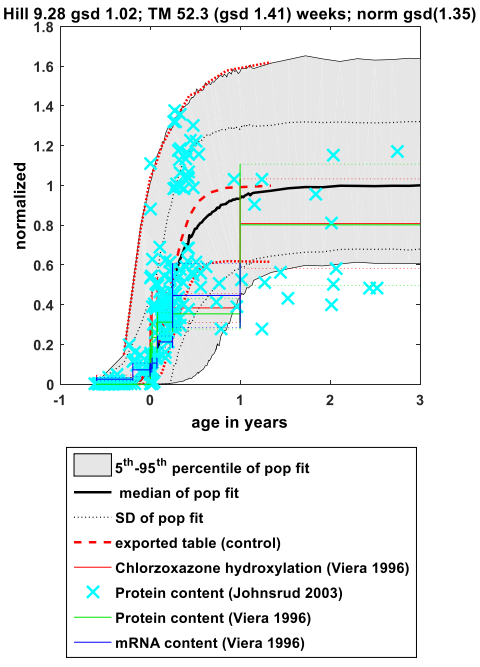
A.7.2 Fit results

A.7.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.7.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

Fit Result:
n = 9.277
$A_{0.5}$ = 52.268

GeoSD n = 1.022
GeoSD $A_{0.5}$ = 1.409
GeoSD adult = 1.350

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.7.2.3 Description for PK-Sim®

Values for CYP2E1 ontogeny are based on information to age dependency of mRNA expression, protein content and chlorzoxazone hydroxylation activity, derived from the papers as mentioned in table A.7.1.

A.7.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP2E1	Standard Deviation
0.48	0.00099	21.07
0.66	0.021	9.07
0.77	0.077	4.85
0.82	0.13	3.67
0.9	0.26	2.51
1.05	0.6	1.63
1.12	0.73	1.47
1.21	0.85	1.39
1.3	0.92	1.33
1.43	0.96	1.32
1.62	0.99	1.33
2.12	1	1.34

A.8 CYP3A4 in liver

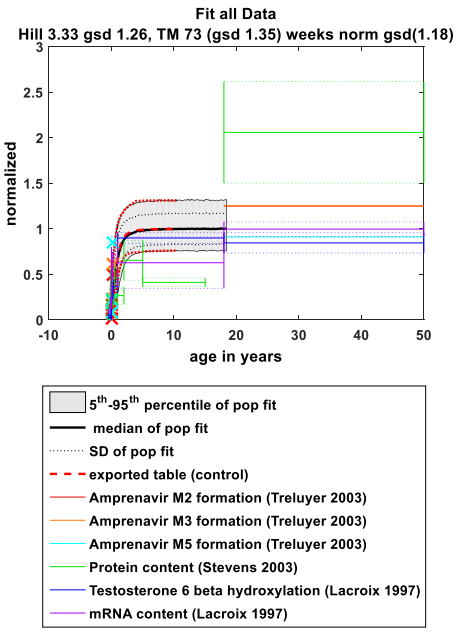
A.8.1 Original data

Reference	Used for fit
Eur J Biochem. 1997 Jul 15;247(2):625-34.; Expression of CYP3A in the human liver--evidence that the shift between CYP3A7 and CYP3A4 occurs immediately after birth.; Lacroix D, Sonnier M, Moncion A, Cheron G, Cresteil T. Digital Object Identifyer (DOI): 10.1111/j.1432-1033.1997.00625.x → Figure 3 (mRNA expression) → Figure 4A-B (Testosterone 6 beta hydroxylation)	 Yes Yes
Drug Metab Dispos. 2003 Mar;31(3):275-81.; Oxidative metabolism of amprenavir in the human liver. Effect of the CYP3A maturation.; Tréluyer JM, Bowers G, Cazali N, Sonnier M, Rey E, Pons G, Cresteil T. Digital Object Identifyer (DOI): 10.1124/dmd.31.3.275 → Table 5 (Amprenavir M2, M3 and M5 and formation)	 Yes
J Pharmacol Exp Ther. 2003 Nov;307(2):573-82. Epub 2003 Sep 15.; Developmental expression of the major human hepatic CYP3A enzymes.; Stevens JC, Hines RN, Gu C, Koukouritaki SB, Manro JR, Tandler PJ, Zaya MJ. Digital Object Identifyer (DOI): 10.1124/jpet.103.054841 → Figure 8 (protein content)	 Yes
Drug Metab Dispos. 2000 Apr;28(4):379-82.; Human cytochrome P450 maximal activities in pediatric versus adult liver.; Blanco JG1, Harrison PL, Evans WE, Relling MV. PMID: 10725303 (DOI not available) → Figure 2 (Midazolam 1' hydroxylation)	 No
Pharmacogenetics. 1997 Dec;7(6):441-52.; Developmental expression of CYP2C and CYP2C-dependent activities in the human liver: in-vivo/in-vitro correlation and inducibility.; Treluyer JM, Gueret G, Cheron G, Sonnier M, Cresteil T. Pubmed ID (PMID): 9429229 (DOI not available) → Figure 4B (Diazepam hydroxylation)	 No

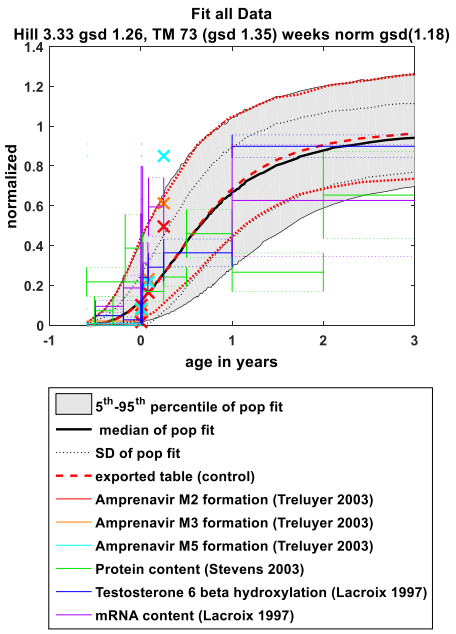
A.8.2 Fit results

A.8.2.1 Figure

Full range-plot



Zoom-in, x-axis



Comment:

The old PK-Sim® version was not fitted using the Hill equation but using spline functions allowing for the 'overshoot' of relative expression compared to adults. According to Edginton et al. 2006 an ontogeny factor of 1.3 was found for children from 1-3 y.

Reassessing in the light of additional data, the references quoted by Edginton et al 2006, such a high expression value for this age range could not be confirmed. The only CYP-mediated process found to yield an ontogeny factor of approx. 1.44 for children aged 3-12 months was reported by Treluyer et al. 1997 for the hydroxylation of diazepam (Fig. 4B). This metabolization, however, is not specific for CYP3A4 but is also mediated by CYP2C19 as reported by Riss et al. 2008 (Acta Neurol Scand. 118(2):69-86). Therefore, the age-dependency of this process was not considered for fitting CYP3A4 ontogeny in liver. Nevertheless, even if this dataset would have been considered, due to the different fit function the new ontogeny profile of CYP3A4 in liver would be significantly different than the old one.

A.8.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

Fit Result:
n = 3.331
$A_{0.5}$ = 73.019

GeoSD n = 1.255
GeoSD $A_{0.5}$ = 1.345
GeoSD adult = 1.182

A.8.2.3 Description for PK-Sim®

Values for CYP3A4 ontogeny in liver are based on information to age dependency of mRNA content, testosterone 6 beta hydroxylation activity, protein content as well as amprenavir metabolite formation, derived from the papers as mentioned in table A.8.1.

A.8.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP3A4	Standard Deviation
0.18	0.001	4.63
0.37	0.011	3.22
0.5	0.03	2.73
0.63	0.064	2.46
0.77	0.12	2.22
0.92	0.2	1.92
1.39	0.49	1.45
1.55	0.58	1.37
1.73	0.67	1.31
1.91	0.74	1.26
2.08	0.79	1.23
2.26	0.83	1.22
2.5	0.87	1.19
2.84	0.91	1.18
3.19	0.94	1.18
3.81	0.97	1.18
4.69	0.98	1.19
6.48	0.99	1.18
11.17	1	1.18

A.9 CYP3A5 in liver

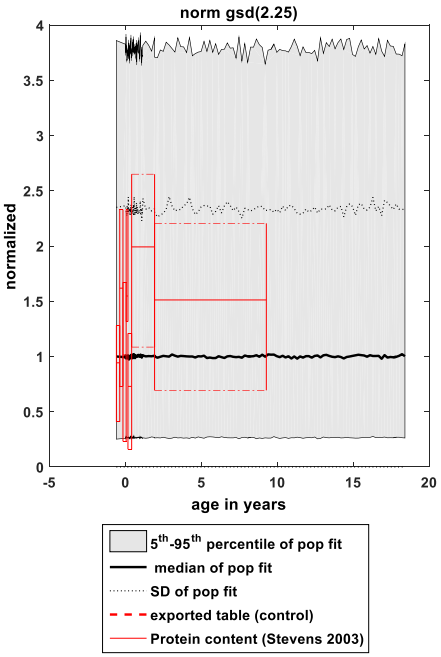
A.9.1 Original data

Reference	Used for fit
J Pharmacol Exp Ther. 2003 Nov;307(2):573-82. Epub 2003 Sep 15.; Developmental expression of the major human hepatic CYP3A enzymes.; Stevens JC, Hines RN, Gu C, Koukouritaki SB, Manro JR, Tandler PJ, Zaya MJ. Digital Object Identifyer (DOI): 10.1124/jpet.103.054841 → Figure 2 (protein content)	yes →Although, no ontogeny visible for this protein.

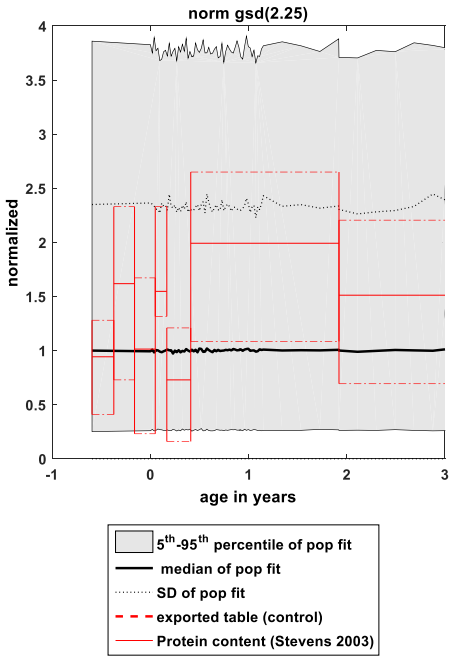
A.9.2 Fit results

A.9.2.1 Figure

Full range-plot



Zoom-in, x-axis



Comment:

Due to the variability of the data, no ontogeny fit for CYP3A5 is possible. The ontogeny function for this enzyme therefore is equal to one for all age groups.

A.9.2.2 Fit function

The ontogeny is equal to 1 for all ages. The geometric standard deviation in activity observed in adults is 2.246.

A.9.2.3 Description for PK-Sim®

CYP3A5 activity in liver is assumed to be at 100 % adult activity at all ages based on the data from Stevens et al. 2003 J Pharmacol Exp Ther 307(2):573-82.

A.9.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP3A5	Standard Deviation
0	1	2.25
0.77	1	2.25

CYP3A7 in liver

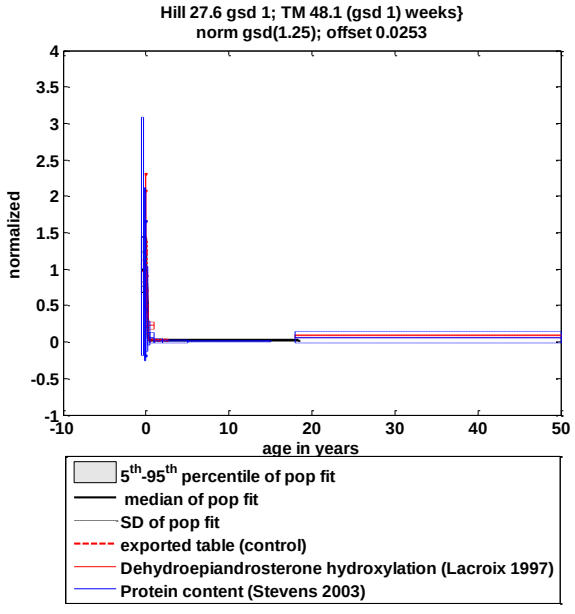
A.9.3 Original data

Reference	Used for fit
Eur J Biochem. 1997 Jul 15;247(2):625-34.; Expression of CYP3A in the human liver--evidence that the shift between CYP3A7 and CYP3A4 occurs immediately after birth.; Lacroix D, Sonnier M, Moncion A, Cheron G, Cresteil T. Digital Object Identifyer (DOI): 10.1111/j.1432-1033.1997.00625.x → Figure 4A-B (Dehydroepiandrosterone hydroxylation)	Yes
J Pharmacol Exp Ther. 2003 Nov;307(2):573-82. Epub 2003 Sep 15.; Developmental expression of the major human hepatic CYP3A enzymes.; Stevens JC, Hines RN, Gu C, Koukouritaki SB, Manro JR, Tandler PJ, Zaya MJ. Digital Object Identifyer (DOI): 10.1124/jpet.103.054841 → Figure 8 (protein content)	Yes

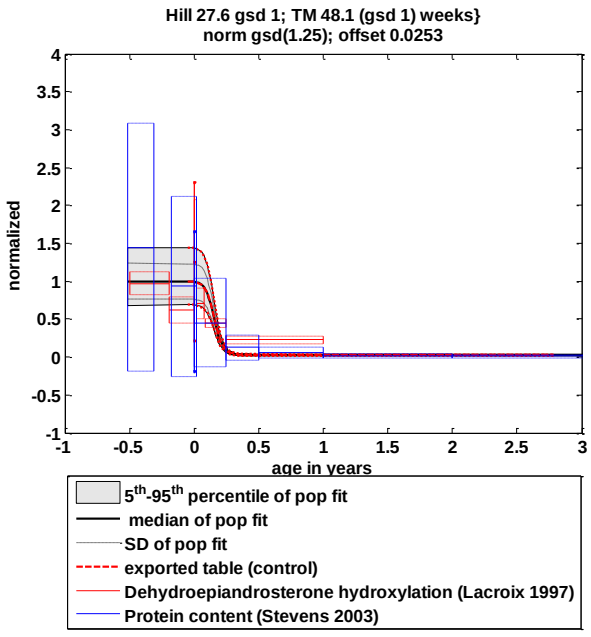
A.9.4 Fit results

A.9.4.1 Figure

Full range-plot



Zoom-in, x-axis



A.9.4.2 Fit function

Fit Result:
n = 27.615
$A_{0.5}$ = 48.051

GeoSD $A_{0.5}$ = 1
GeoSD n = 1
GeoSD adult = 1.254
Offset = 0.0253

$A = 1-PMA^n/(A_{0.5}{}^n + PMA^n) + offset$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults
Offset = activity in adults

A.9.4.3 Description for PK-Sim®

Values for CYP3A7 ontogeny are based on information to age dependency of protein content and dehydroepiandrosterone hydroxylation activity, derived from the papers as mentioned in table A.10.1.

A.9.4.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP3A7	Standard Deviation
0.72	1	1.25
0.77	0.99	1.25
0.83	0.95	1.26
0.87	0.85	1.25
0.9	0.71	1.25
0.95	0.36	1.25
0.98	0.2	1.26
1.02	0.078	1.25
1.05	0.054	1.26
1.12	0.03	1.25
1.83	0.025	1.26
3.36	0.025	1.25
3.41	0.025	1.25
3.56	0.025	1.26

4.4 UGT enzymes in liver

A.1 UGT1A1 in liver

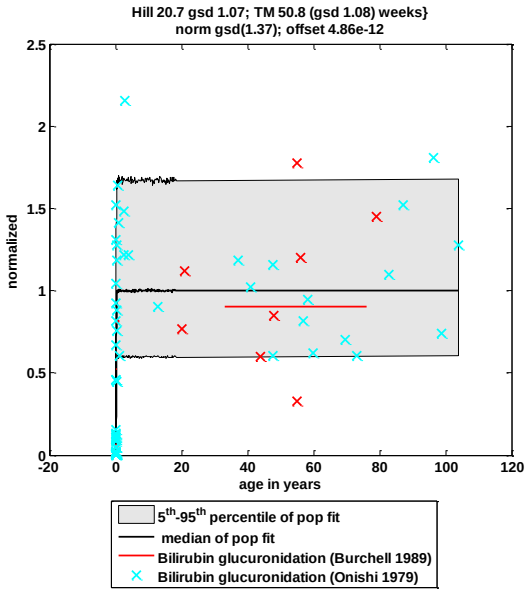
A.1.1 Original data

Reference	Used for fit
Dev Pharmacol Ther. 1989;13(2-4):70-7.; Development of human liver UDP-glucuronosyltransferases.; Burchell B, Coughtrie M, Jackson M, Harding D, Fournel-Gigleux S, Leakey J, Hume R. Pubmed ID (PMID): 2515047 (DOI not available) → Table 1 (Bilirubin glucuronidation) → Table 2 (Bilirubin glucuronidation)	 No Yes
Drug Metab Dispos. 2011 May;39(5):912-9. doi: 10.1124/dmd.110.037192. Epub 2011 Jan 25.; The development of UDP-glucuronosyltransferases 1A1 and 1A6 in the pediatric liver.; Miyagi SJ, Collier AC. Digital Object Identifyer (DOI): 10.1124/dmd.110.037192 → Figure 1A (Bilirubin glucuronidation)	 No
Gut. 2002 Feb;50(2):259-65.; Developmental aspects of human hepatic drug glucuronidation in young children and adults.; Strassburg CP, Strassburg A, Kneip S, Barut A, Tukey RH, Rodeck B, Manns MP. Digital Object Identifyer (DOI): 10.1136/gut.50.2.259 → Figure 2 (mRNA expression)	 No
Biochem J. 1979 Dec 15;184(3):705-7.; Postnatal development of uridine diphosphate glucuronyltransferase activity towards bilirubin and 2-aminophenol in human liver.; Onishi S, Kawade N, Itoh S, Isobe K, Sugiyama S. Digital Object Identifyer (DOI): 10.1042/bj1840705 → Figure 1 (Bilirubin glucuronidation)	 Yes

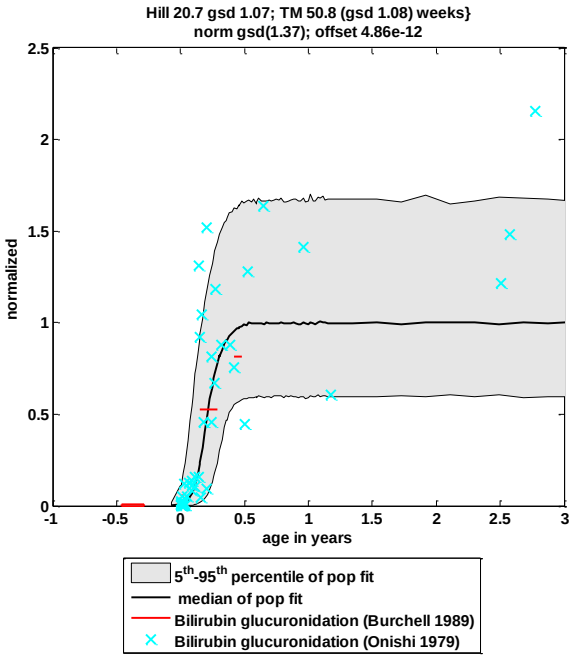
A.1.2 Fit results

A.1.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.1.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

Fit Result:
n = 20.670
$A_{0.5}$ = 50.754

GeoSD n = 1.066
GeoSD $A_{0.5}$ = 1.084
GeoSD adult = 1.367

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.1.2.3 Description for PK-Sim®

Values for UGT1A1 ontogeny are based on information to age dependency of mRNA expression, and bilirubin glucuronidation activity, derived from the papers as mentioned in table A.2.1.

A.1.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for UGT1A1	Standard Deviation
0.698271871	0.000979058	2.552011728
0.769230783	0.007233677	2.38158989
0.858116925	0.064657465	2.010226488

0.890073895	0.130265653	1.816496134
0.919260323	0.224505559	1.637136698
1.029093981	0.755379558	1.228398442
1.075121403	0.884206295	1.176906228
1.128819823	0.954406977	1.152582049
1.226630092	0.991511762	1.140699267
1.361833453	0.999002397	1.138464808

A.2 UGT1A4 in liver

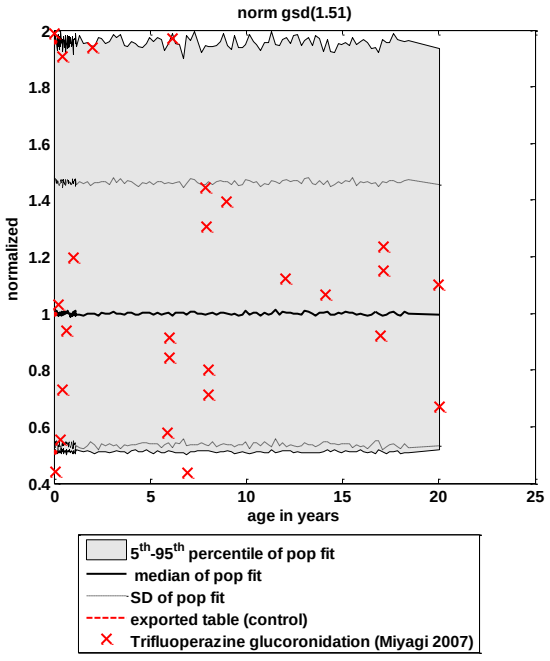
A.2.1 Original data

Reference	Used for fit
Drug Metab Dispos. 2007 Sep;35(9):1587-92. Epub 2007 Jun 7.; Pediatric development of glucuronidation: the ontogeny of hepatic UGT1A4.; Miyagi SJ, Collier AC. Digital Object Identifyer (DOI): 10.1124/dmd.107.015214 → Figure 3A (Trifluoperazine glucoronidation)	Yes → although no ontogeny visible for this protein

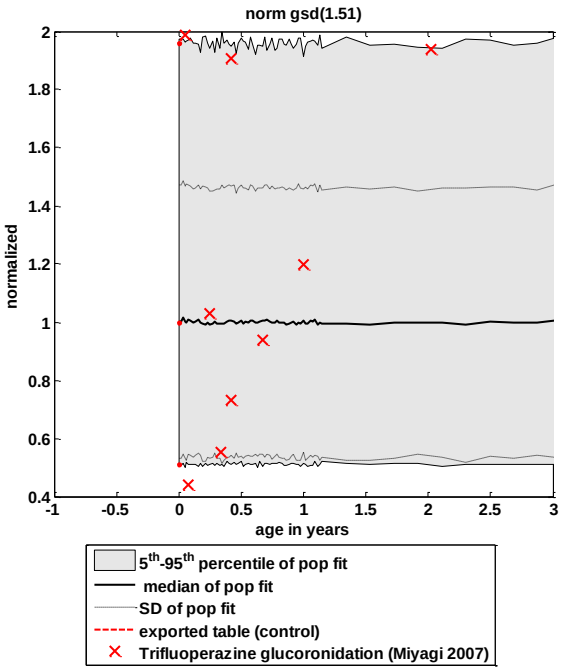
A.2.2 Fit results

A.2.2.1 Figure

Full range-plot



Zoom-in, x-axis



Comment:

Due to the variability of the data, no ontogeny fit using the Hill equation for UGT1A4 is possible. The ontogeny function for this enzyme therefore is equal to one for all age groups.

The curve shown in Miyagi et al. 2007 is a result of modeling hepatic UGT1A4 liver clearance assuming a constant rate of development. It therefore does not represent a true fit.

A.2.2.2 Fit function

The ontogeny is equal to 1 for all ages. The geometric standard deviation in activity observed in adults is 1.506.

A.2.2.3 Description for PK-Sim®

UGT1A4 activity in liver is assumed to be at 100 % adult activity at all ages based on the data from Miyagi 2007 Drug Metab Dispos 35(9):1587-1592.

A.2.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for UGT1A4	Standard Deviation
0	1	1.51
0.77	1	1.51

A.3 UGT1A6 in liver

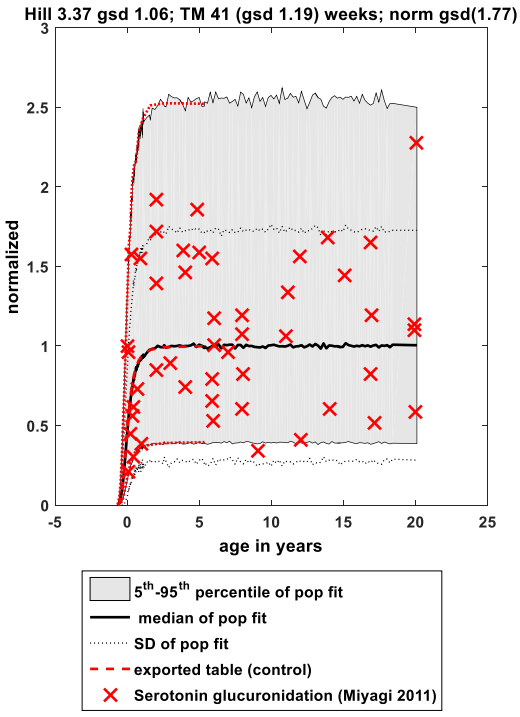
A.3.1 Original data

Reference	Used for fit
Drug Metab Dispos. 2011 May;39(5):912-9. doi: 10.1124/dmd.110.037192. Epub 2011 Jan 25.; The development of UDP-glucuronosyltransferases 1A1 and 1A6 in the pediatric liver.; Miyagi SJ, Collier AC. Digital Object Identifyer (DOI): 10.1124/dmd.110.037192 → Figure 2A (Serotonin glucuronidation)	Yes

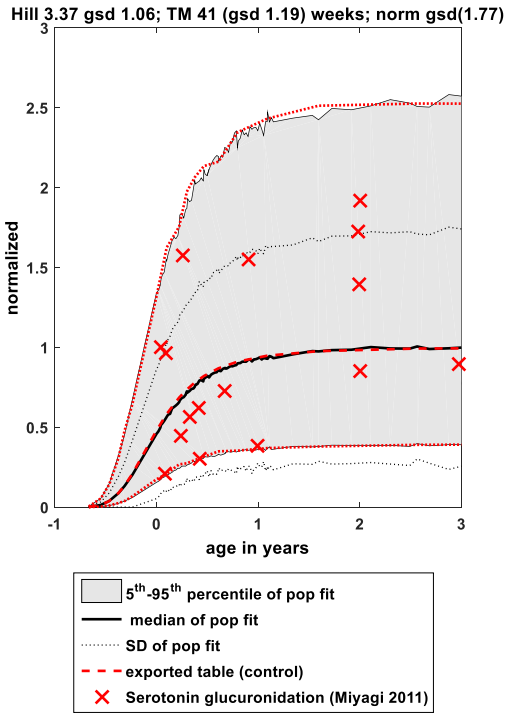
A.3.2 Fit results

A.3.2.1 Figure

Full range-plot



Zoom-in, x-axis



Comment:

The old PK-Sim fit was based on in vivo data of paracetamol glucuruonidation, i.e. paracetamol glucuronide to sulfate ratios (Edginton et al. 2006). The fit using in vitro serotonin glucuronidation data, however, shows a similar representation of UGT1A6 ontogeny.

A.3.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:

PK-Sim® Ontogeny Database

Fit Result:
n = 3.369
$A_{0.5}$ = 41.030

GeoSD n = 1.064
GeoSD $A_{0.5}$ = 1.192
GeoSD adult = 1.769

PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.3.2.3 Description for PK-Sim®

Values for UGT1A6 ontogeny are based on information to age dependency of serotonin glucuronidation as reported by Miyagi 2011 Drug Metab Dispos 39(5): 912-919.

A.3.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for UGT1A6	Standard Deviation
0.1	0.001	2.49
0.22	0.013	2.36
0.3	0.038	2.23
0.39	0.088	2.14
0.45	0.13	2.16
0.53	0.21	1.99
0.77	0.48	1.85
0.87	0.58	1.87
0.99	0.69	1.77
1.07	0.74	1.82
1.15	0.78	1.81
1.24	0.82	1.79
1.38	0.87	1.74
1.57	0.91	1.78
1.86	0.95	1.77
2.37	0.98	1.78
3.34	0.99	1.76
6.13	1	1.76

A.4 UGT1A9 in liver

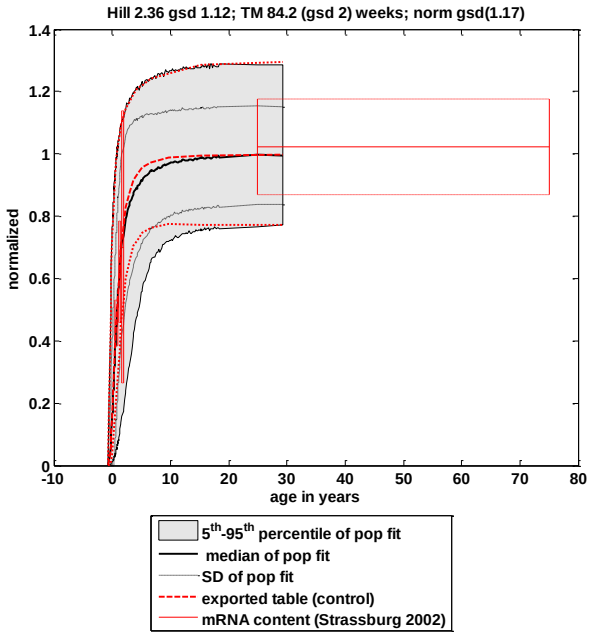
A.4.1 Original data

Reference	Used for fit
Gut. 2002 Feb;50(2):259-65.; Developmental aspects of human hepatic drug glucuronidation in young children and adults.; Strassburg CP, Strassburg A, Kneip S, Barut A, Tukey RH, Rodeck B, Manns MP. Digital Object Identifyer (DOI): 10.1136/gut.50.2.259 → Figure 2 (mRNA expression)	Yes

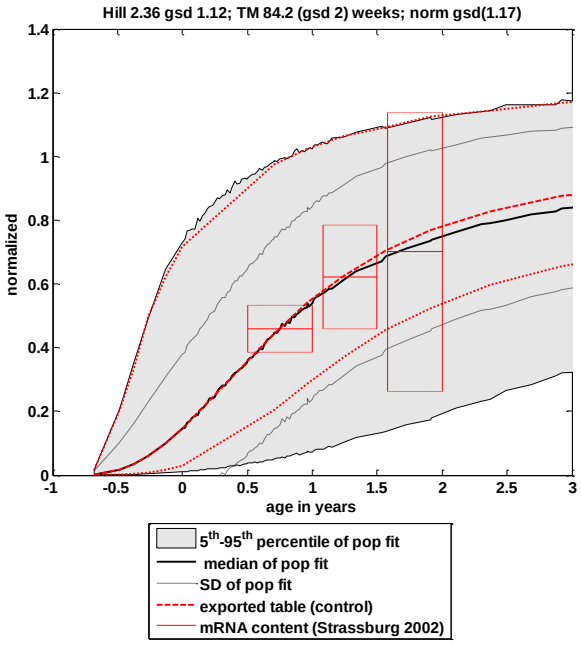
A.4.2 Fit results

A.4.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.4.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

Fit Result:
n = 2.362
$A_{0.5}$ = 84.220

GeoSD n = 1.122
GeoSD $A_{0.5}$ = 1.997
GeoSD adult = 1.169

A.4.2.3 Description for PK-Sim®

Values for UGT1A9 ontogeny are based on information to age dependency of mRNA expression as reported by Strassburg et al. 2002 Gut 50:259–265.

A.4.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for UGT1A9	Standard Deviation
0.09	0.00099	5.48
0.29	0.016	4.63
0.4	0.035	3.99
0.5	0.059	3.64
0.64	0.1	3.03
0.77	0.15	2.62
1.47	0.44	1.61
1.74	0.54	1.47
2.01	0.63	1.38
2.34	0.7	1.3
2.69	0.77	1.26
3.14	0.83	1.22
3.7	0.88	1.19
4.47	0.92	1.17
5.03	0.94	1.17
5.95	0.96	1.16
7.45	0.97	1.16
10.49	0.99	1.16
16.12	1	1.17

A.5 UGT2B4 in liver

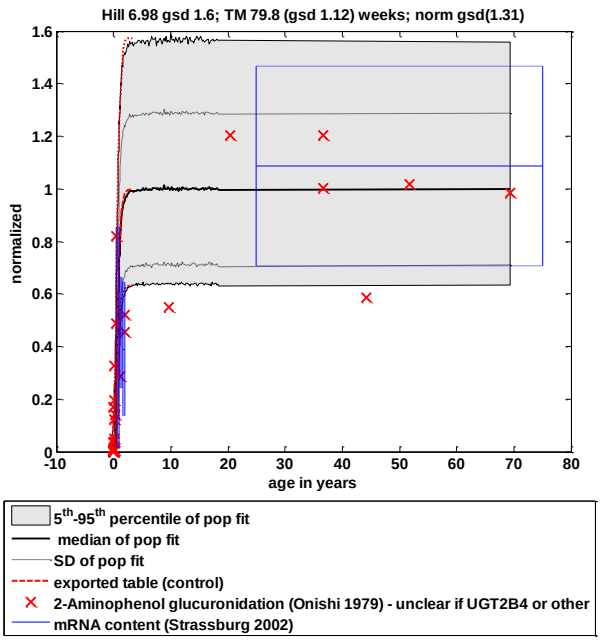
A.5.1 Original data

Reference	Used for fit
Biochem J. 1979 Dec 15;184(3):705-7.; Postnatal development of uridine diphosphate glucuronyltransferase activity towards bilirubin and 2-aminophenol in human liver.; Onishi S, Kawade N, Itoh S, Isobe K, Sugiyama S. Digital Object Identifyer (DOI): 10.1042/bj1840705 → Figure 2 (2-aminophenol glucuronidation)	Yes
Gut. 2002 Feb;50(2):259-65.; Developmental aspects of human hepatic drug glucuronidation in young children and adults.; Strassburg CP, Strassburg A, Kneip S, Barut A, Tukey RH, Rodeck B, Manns MP. Digital Object Identifyer (DOI): 10.1136/gut.50.2.259 → Figure 2 (mRNA expression)	Yes

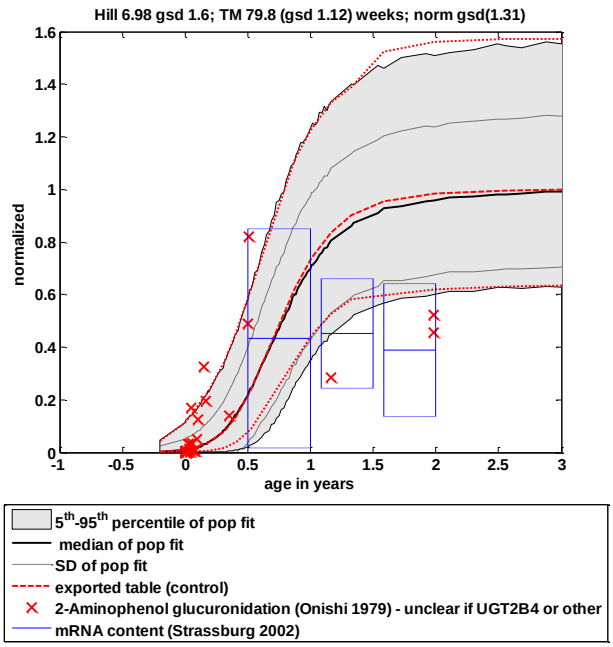
A.5.2 Fit results

A.5.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.5.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

Fit Result:
n = 6.983
$A_{0.5}$ = 79.785

GeoSD n = 1.596
GeoSD $A_{0.5}$ = 1.125
GeoSD adult = 1.314

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

A.5.2.3 Description for PK-Sim®

Values for UGT2B4 ontogeny are based on information to age dependency of 2-aminophenol Glucuronidation, and mRNA expression, derived from the papers as mentioned in table A.5.1.

A.5.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for UGT2B7	Standard Deviation
0.479999989	0.00099	17.12999916
0.660000026	0.0073	12.78999996
0.769999981	0.02	8.729999542
0.910000026	0.057	5.679999828
1.029999971	0.119999997	3.970000029
1.149999976	0.219999999	2.960000038
1.24000001	0.310000002	2.480000019
1.5	0.620000005	1.769999981
1.639999986	0.74000001	1.669999957
1.75999999	0.819999993	1.610000014
1.889999986	0.879999995	1.570000052
2.069999933	0.930000007	1.539999962
2.349999905	0.970000029	1.519999981
2.859999895	0.99000001	1.559999943
4.010000229	1	1.590000033

A.6 UGT2B7 in liver

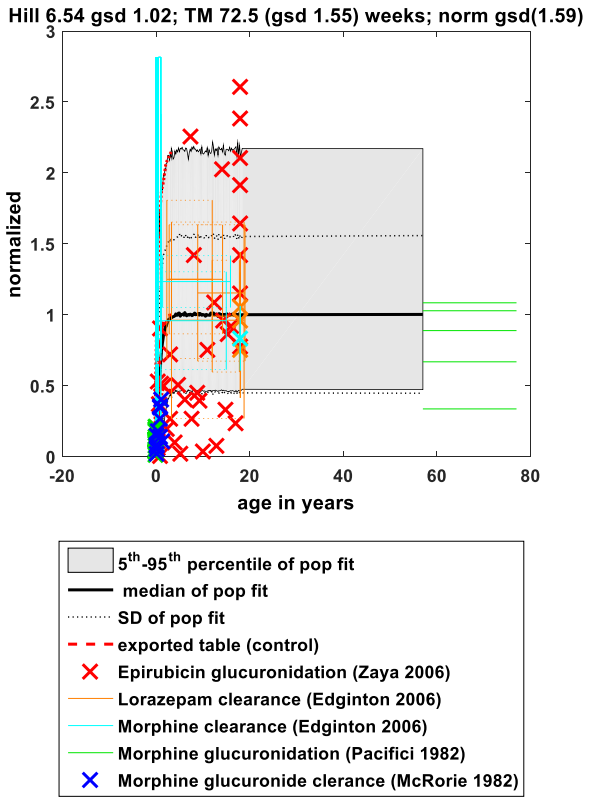
A.6.1 Original data

Reference	Used for fit
Gut. 2002 Feb;50(2):259-65.; Developmental aspects of human hepatic drug glucuronidation in young children and adults.; Strassburg CP, Strassburg A, Kneip S, Barut A, Tukey RH, Rodeck B, Manns MP. Digital Object Identifyer (DOI): 10.1136/gut.50.2.259 → Figure 2 (mRNA expression)	No
Drug Metab Dispos. 2006 Dec;34(12):2097-101.; Epub 2006 Sep 19. Epirubicin glucuronidation and UGT2B7 developmental expression.; Zaya MJ1, Hines RN, Stevens JC. Digital Object Identifyer (DOI): 10.1124/dmd.106.011387 → Figure 6 A (Epirubicin glucuronidation)	Yes
Eur J Clin Pharmacol. 1982;22(6):553-8.; Morphine glucuronidation in human fetal and adult liver.; Pacifici GM, Säwe J, Kager L, Rane A. Digital Object Identifyer (DOI): 10.1007/BF00609630 → Figure 2 (Morphine glucuronidation)	Yes
Am J Dis Child. 1992 Aug;146(8):972-6.; The maturation of morphine clearance and metabolism.; McRorie TI, Lynn AM, Nespeca MK, Opheim KE, Slattery JT. Digital Object Identifyer (DOI): 10.1001/archpedi.1992.02160200094036 → Figure 2 (Morphine glucuronide clearance)	Yes
Clin Pharmacokinet. 2006;45(10):1013-34.; Development and evaluation of a generic physiologically based pharmacokinetic model for children.; Edginton AN, Schmitt W, Willmann S. Digital Object Identifyer (DOI): 10.2165/00003088-200645100-00005 →Table 3 (Lorazepam and morphine clearance)	Yes

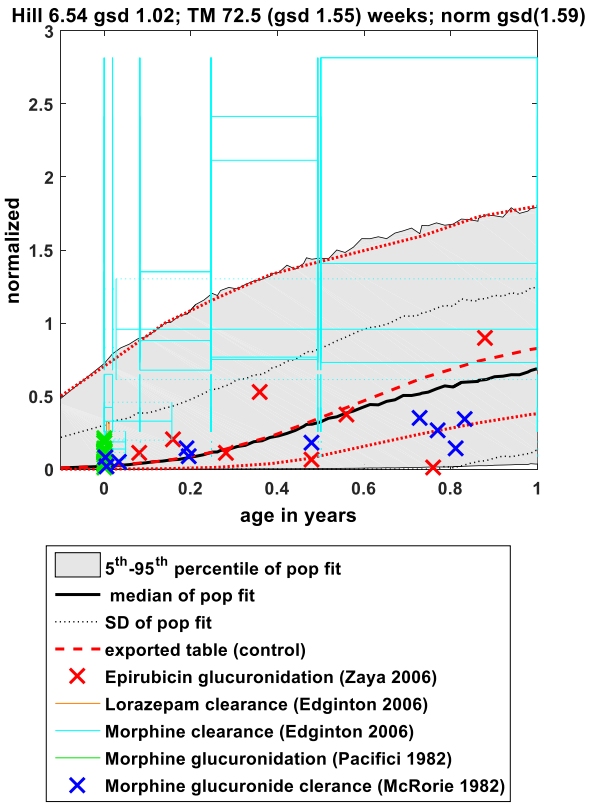
A.6.2 Fit results

A.6.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.6.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation
GeoSD adult = geometric standard deviation in activity observed in adults

Fit Result:
n = 6.543
$A_{0.5}$ = 72.533

GeoSD n = 1.023
GeoSD $A_{0.5}$ = 1.554
GeoSD adult = 1.595

A.6.2.3 Description for PK-Sim®

Values for UGT2B7 ontogeny are based on information to age dependency morphine glucuronidation activity, epirubicin glucuronidation activity, as well as in vivo morphine and lorazepam glucuronide clearance, derived from the papers as mentioned in table A.6.1.

A.6.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for UGT2B7	Standard Deviation
0.479999989	0.00099	17.12999916
0.660000026	0.0073	12.78999996

0.769999981	0.02	8.729999542
0.910000026	0.057	5.679999828
1.029999971	0.119999997	3.970000029
1.149999976	0.219999999	2.960000038
1.24000001	0.310000002	2.480000019
1.5	0.620000005	1.769999981
1.639999986	0.74000001	1.669999957
1.75999999	0.819999993	1.610000014
1.889999986	0.879999995	1.570000052
2.069999933	0.930000007	1.539999962
2.349999905	0.970000029	1.519999981
2.859999895	0.99000001	1.559999943
4.010000229	1	1.590000033

4.5 CYP Enzymes in the intestine

A.1 CYP3A4 in duodenum

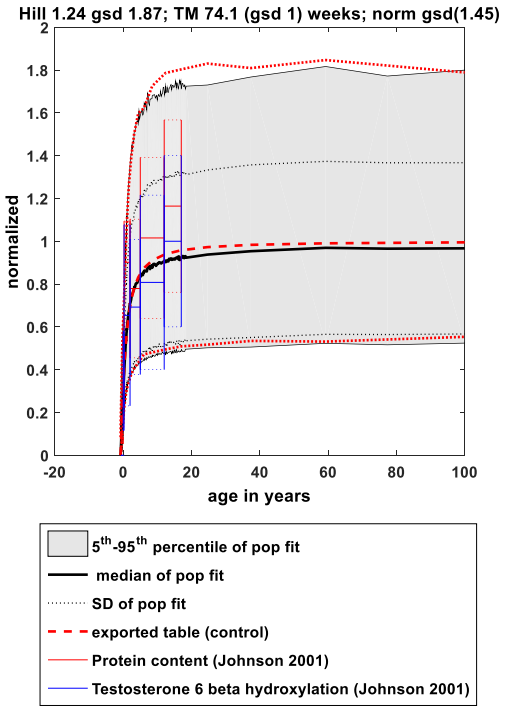
A.1.1 Original data

Reference	Used for fit
Br J Clin Pharmacol. 2001 May;51(5):451-60.; Enterocytic CYP3A4 in a paediatric population: developmental changes and the effect of coeliac disease and cystic fibrosis.; Johnson TN, Tanner MS, Taylor CJ, Tucker GT. Digital Object Identifyer (DOI): 10.1046/j.1365-2125.2001.01370.x → Figure 2 (Protein content and Testosterone 6 beta hydroxylation)	Yes

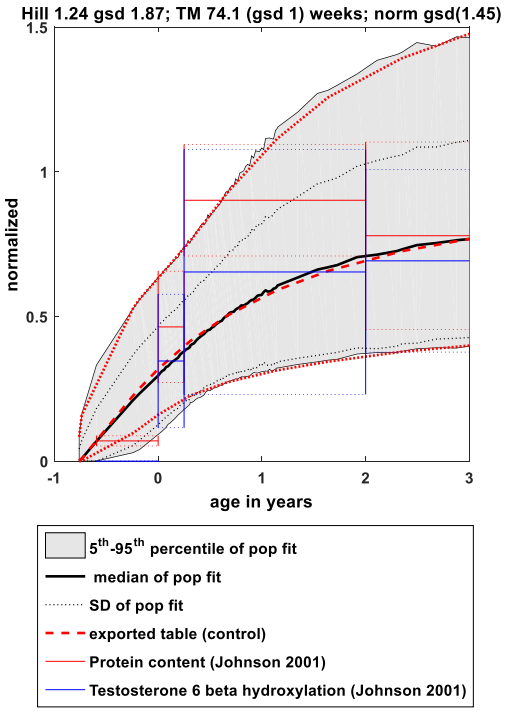
A.1.2 Fit results

A.1.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.1.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation

Fit Result:
n = 1.237
$A_{0.5}$ = 74.055

GeoSD n = 1.872
GeoSD $A_{0.5}$ = 1.000
GeoSD adult = 1.451

GeoSD adult = geometric standard deviation in activity observed in adults

A.1.2.3 Description for PK-Sim®

Values for CYP3A4 ontogeny in intestine are based on information to age dependency of protein content and testosterone 6 beta hydroxylation activity as described by Johnson et al. 2001 Br J Clin Pharmacol. 51(5):451-60.

A.1.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CYP3A4	Standard Deviation
0.01	0.0011	13.99
0.03	0.0086	5.87
0.53	0.23	1.67
0.59	0.25	1.63
0.77	0.32	1.52
1.08	0.41	1.45
1.44	0.5	1.45
1.92	0.59	1.47
2.41	0.66	1.48
3.13	0.73	1.49
3.97	0.78	1.49
5.17	0.83	1.49
6.87	0.87	1.46
9.29	0.91	1.47
13.18	0.94	1.48
17.8	0.96	1.47
25.51	0.97	1.47
38.25	0.98	1.45
60.21	0.99	1.46
101.04	1	1.43

4.6 Other enzymes

A.1 Carboxylesterase 1 (CES1)

The CES-1 ontogeny for PK-Sim is based on the publication of Boberg et al. 2017. In the publication, an equation for the mean ontogeny is already given. Aim of the below described steps was the re-normalization of the ontogeny to reach a relative expression level of 1 for adults and the calculation of variability in dependence of age.

A.1.1 Original data

Reference	Used for fit
Boberg M, Vrana M, Mehrotra A, Pearce RE, Gaedigk A, Bhatt DK, Leeder JS, Prasad B. Age-Dependent Absolute Abundance of Hepatic Carboxylesterases (CES1 and CES2) by LC-MS/MS Proteomics: Application to PBPK Modeling of Oseltamivir In Vivo Pharmacokinetics in Infants. Drug Metab Dispos. 2017 Feb;45(2):216-223. doi: 10.1124/dmd.116.072652. Epub 2016 Nov 28. PMID: 27895113; PMCID: PMC5267516. → Table 1 & 2	See below

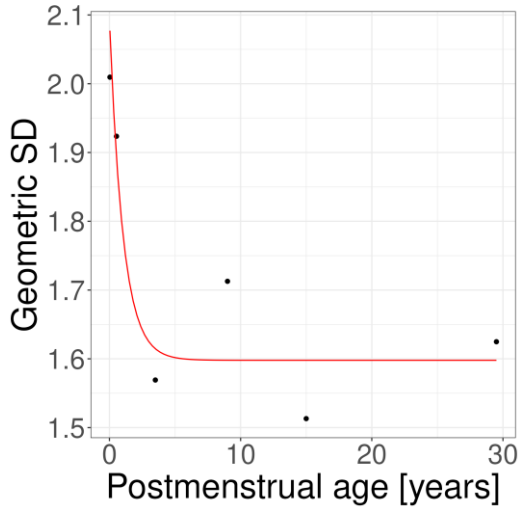
A.1.2 Ontogeny results

A.1.2.1 Data preparation

- Extraction of age group specific microsomal and cytosolic CES-1 concentrations from Table 2 (Boberg et al. 2017)
- Transformation of abundance values from pmol/mg total protein to pmol/mL of liver tissue using reported values of milligrams of microsomal and cytosolic proteins per gram of liver tissue (39.8 and 80.7 mg/ml, respectively; Boberg et al. 2017)
- Estimation of geometric means and geometric SDs from given arithmetic means and arithmetic SDs
    - geomean= mean^2/sqrt(SD^2+mean^2)
    - geosd = exp(sqrt(log(SD^2/mean^2+1)))
- Sum up values for microsomal and cytosolic protein concentrations to get total abundance for each age group
    - geomean_liver = sum(geomean_microsomal + geomean_cytosol)
    - geoSD_liver = exp(sqrt(((log(geoSD_microsomal)^2+log(geoSD_cytosol)^2))/2))

A.1.2.2 Variability fit

- Fit of an exponential decay model on the age-group dependent geoSDs
    - Model: exp(-k*Age)*(Onset-Offset)+Offset
    - Fit results:
        - K = 0.969
        - Onset = 2.095
        - Offset = 1.598

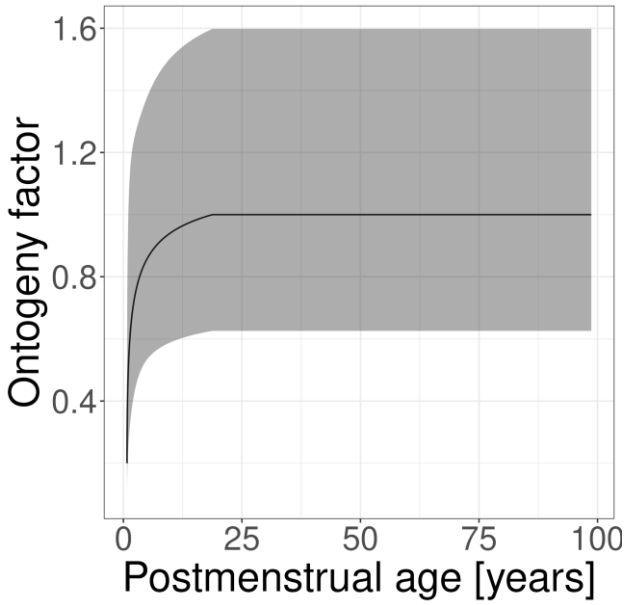


A.1.2.3 Re-normalization and ontogeny generation

- The geomean of the final ontogeny was calculated as follows:
  - Geomean of ontogeny factor calculated according to equation given in Boberg et al 2017 (Table 1)

Ontogeny equation[a]	$F = \left(\frac{Adult_{max} - F_{birth}}{Age50^n + Age^n}\right) \times Age^n + F_{birth}$
CES1 ontogeny parameters[b]	F_{birth}, $Adult_{max}$, Age_{50}, and n: 0.20, 1, 1.10, and 0.56, respectively[b]

  - There is no information given in the publication regarding ontogeny in premature infants. Therefore, premature age ranges are not included in the final ontogeny.
  - Since the ontogeny never reaches 1 and individuals at 18 are only at ~0.86 the $Adult_{max}$ value was set to 1.167231 (calculated by setting F to 1 and age to 18 and then resolving the equation for $Adult_{max}$) so that the ontogeny factor at age 18 is 1. All ages above were set to the same value as for 18year old individuals.
- The geoSD of the final ontogeny was calculated for each age step using the previously fitted equation.
- Final ontogeny:



A.1.2.4 Description for PK-Sim®

Values for CES1 ontogeny are based on information to age dependency of protein content as described by Boberg et al. 2017.

A.1.2.5 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for CES1 (Geometric Mean)	Geometric Standard Deviation
0.76659822	0.2	2.094629148
0.769336071	0.232541461	2.093313636
0.772073922	0.247221396	2.092001606
0.774811773	0.258530006	2.090693052
0.777549624	0.268041588	2.089387963
0.780287474	0.276382962	2.08808633
0.783025325	0.283881631	2.086788144
0.785763176	0.290734936	2.085493396
0.788501027	0.297072945	2.084202077
0.791238877	0.302986881	2.082914178
0.793976728	0.308543719	2.08162969
0.796714579	0.313794388	2.080348603
0.79945243	0.318778707	2.079070908
0.802190281	0.323528529	2.077796598
0.804928131	0.328069809	2.076525662
0.807665982	0.332424041	2.075258092
0.810403833	0.336609265	2.073993878
0.813141684	0.340640798	2.072733013
0.815879535	0.344531783	2.071475487
0.818617385	0.348293596	2.070221291
0.821355236	0.351936165	2.068970416
0.824093087	0.355468216	2.067722854
0.826830938	0.358897466	2.066478596
0.829568789	0.362230782	2.065237633
0.832306639	0.365474305	2.063999957
0.83504449	0.368633555	2.062765558
0.837782341	0.371713516	2.061534428
0.840520192	0.374718702	2.060306559
0.843258042	0.377653223	2.059081941
0.845995893	0.38052083	2.057860566
0.848733744	0.383324957	2.056642426
0.85152974	0.386126401	2.055401745
0.855884272	0.390371352	2.053476166
0.860462066	0.394688986	2.051460598
0.865274568	0.399079451	2.049351303
0.870333813	0.403542839	2.047144437
0.87565245	0.408079189	2.044836054
0.88124378	0.412688484	2.042422105
0.887121784	0.417370647	2.039898446
0.893301159	0.422125541	2.037260842
0.899797357	0.426952966	2.034504974
0.906626623	0.431852659	2.031626445
0.913806033	0.436824291	2.02862079

0.921353539	0.441867467	2.025483486
0.929288014	0.446981723	2.022209965
0.937629298	0.452166525	2.018795623
0.946398249	0.457421269	2.015235843
0.955616793	0.462745278	2.011526006
0.965307983	0.468137803	2.007661512
0.97549605	0.47359802	2.003637803
0.986206471	0.479125033	1.999450387
0.997466027	0.484717867	1.995094864
1.009302872	0.490375476	1.990566956
1.021746606	0.496096733	1.985862538
1.034828343	0.501880441	1.980977675
1.048580796	0.507725322	1.975908659
1.063038351	0.513630026	1.970652051
1.078237162	0.519593125	1.965204726
1.094215232	0.525613119	1.959563917
1.111012516	0.531688433	1.953727273
1.128671014	0.537817419	1.947692906
1.147234883	0.543998357	1.941459453
1.166750542	0.550229458	1.935026133
1.18726679	0.556508864	1.928392811
1.208834929	0.562834648	1.921560064
1.23150889	0.569204819	1.914529245
1.25534537	0.575617325	1.907302555
1.280403972	0.58207005	1.899883107
1.306747356	0.588560821	1.892275001
1.334441395	0.595087409	1.884483388
1.363555337	0.601647534	1.876514539
1.394161983	0.608238863	1.868375907
1.426337865	0.614859018	1.860076187
1.46016344	0.621505578	1.851625374
1.495723289	0.628176081	1.8430348
1.533106331	0.634868027	1.834317183
1.572406042	0.641578886	1.825486646
1.613720692	0.648306096	1.816558737
1.65715359	0.655047073	1.807550425
1.70281334	0.661799207	1.798480085
1.750814116	0.668559875	1.789367464
1.801275944	0.675326439	1.780233622
1.854325005	0.68209625	1.771100856
1.91009395	0.688866657	1.761992594
1.96872223	0.695635006	1.752933271
2.030356446	0.702398647	1.743948169
2.095150715	0.709154938	1.735063238
2.163267059	0.715901245	1.726304883
2.234875801	0.722634955	1.717699731
2.310156002	0.729353471	1.709274366
2.389295902	0.736054221	1.701055044
2.472493392	0.742734658	1.693067394
2.559956507	0.74939227	1.685336089
2.651903953	0.756024578	1.677884523

2.748565645	0.762629142	1.670734473
2.850183288	0.769203563	1.663905768
2.957010978	0.775745488	1.657415968
3.069315842	0.782252613	1.651280066
3.187378699	0.788722685	1.645510213
3.311494768	0.795153506	1.640115489
3.441974404	0.801542936	1.635101713
3.579143875	0.807888891	1.630471317
3.723346174	0.814189354	1.626223281
3.874941883	0.820442368	1.62235313
4.034310071	0.826646045	1.618853021
4.20184924	0.832798565	1.615711886
4.377978326	0.838898175	1.612915668
4.563137743	0.844943196	1.610447612
4.757790487	0.85093202	1.608288632
4.962423291	0.856863113	1.60641772
5.177547842	0.862735015	1.604812408
5.403702065	0.86854634	1.603449248
5.641451463	0.874295779	1.602304303
5.891390534	0.879982097	1.601353633
6.154144254	0.885604136	1.600573756
6.430369646	0.891160815	1.599942064
6.720757417	0.896651126	1.599437185
7.026033687	0.902074138	1.599039288
7.346961806	0.907428995	1.598730294
7.684344262	0.912714914	1.598494033
8.039024686	0.917931185	1.598316306
8.411889964	0.923077173	1.598184882
8.803872454	0.928152311	1.598089432
9.215952316	0.933156106	1.598021409
9.649159964	0.93808813	1.597973888
10.10457864	0.942948026	1.597941376
10.58334714	0.947735501	1.597919616
11.08666262	0.952450326	1.597905386
11.61578363	0.957092338	1.597896303
12.17203326	0.961661432	1.597890651
12.75680242	0.966157565	1.597887228
13.37155334	0.97058075	1.597885212
14.01782321	0.974931056	1.597884059
14.69722804	0.979208606	1.59788342
15.4114667	0.983413576	1.597883078
16.16232517	0.987546192	1.5978829
16.95168097	0.991606725	1.597882811
17.78150791	0.995595496	1.597882769
18.65388098	0.999512869	1.597882749
18.76659822	1	1.597882748
19.57098158	1	1.597882748
20.53510293	1	1.597882748
21.54865584	1	1.597882748
22.61417472	1	1.597882748
23.73432392	1	1.597882748

24.9119044	1	1.597882748
26.14986072	1	1.597882748
27.45128842	1	1.597882748
28.81944174	1	1.597882748
30.25774178	1	1.597882748
31.76978505	1	1.597882748
33.35935242	1	1.597882748
35.03041867	1	1.597882748
36.78716231	1	1.597882748
38.63397612	1	1.597882748
40.5754781	1	1.597882748
42.61652302	1	1.597882748
44.76221455	1	1.597882748
47.01791803	1	1.597882748
49.38927391	1	1.597882748
51.8822118	1	1.597882748
54.50296535	1	1.597882748
57.25808781	1	1.597882748
60.15446841	1	1.597882748
63.19934963	1	1.597882748
66.40034524	1	1.597882748
69.76545941	1	1.597882748
73.30310667	1	1.597882748
77.02213299	1	1.597882748
80.93183786	1	1.597882748
85.04199759	1	1.597882748
89.36288971	1	1.597882748
93.90531871	1	1.597882748
98.68064303	1	1.597882748

4.7 Transporter

A.1 P-glycoprotein 1 (P-gp)

The renal ontogeny of P-gp was published by Hunt et al. 2024. This ontogeny goes back to transporter expression measurements from human postmortem renal cortical tissue samples Cheung et al. 2019. The ontogeny of the renal P-gp transporter was applied to all other P-gp expressing organs, mainly, liver and intestine.

A.1.1 Original data

Reference	Used for fit
Hunt JP, Dubinsky S, McKnite AM, Cheung KWK, van Groen BD, Giacomini KM, de Wildt SN, Edginton AN, Watt KM. Maximum likelihood estimation of renal transporter ontogeny profiles for pediatric PBPK modeling. CPT Pharmacometrics Syst Pharmacol. 2024 Apr;13(4):576-588. doi: 10.1002/psp4.13102. Epub 2024 Jan 10. PMID: 38156758; PMCID: PMC11015082. Table S4. P-gp Ontogeny Profiles “Lognormally Distributed”	Published ontogeny values
Cheung KWK, van Groen BD, Spaans E, van Borselen MD, de Bruijn ACJM, Simons-Oosterhuis Y, Tibboel D, Samsom JN, Verdijk RM, Smeets B, Zhang L, Huang SM, Giacomini KM, de Wildt SN. A Comprehensive Analysis of Ontogeny of Renal Drug Transporters: mRNA Analyses, Quantitative Proteomics, and Localization. Clin Pharmacol Ther. 2019 Nov;106(5):1083-1092. doi: 10.1002/cpt.1516. Epub 2019 Jul 3. PMID: 31127606; PMCID: PMC6777991.	Data used in ontogeny function generation in Hunt et al. 2024

A.1.2 Ontogeny results

A.1.2.1 Figure

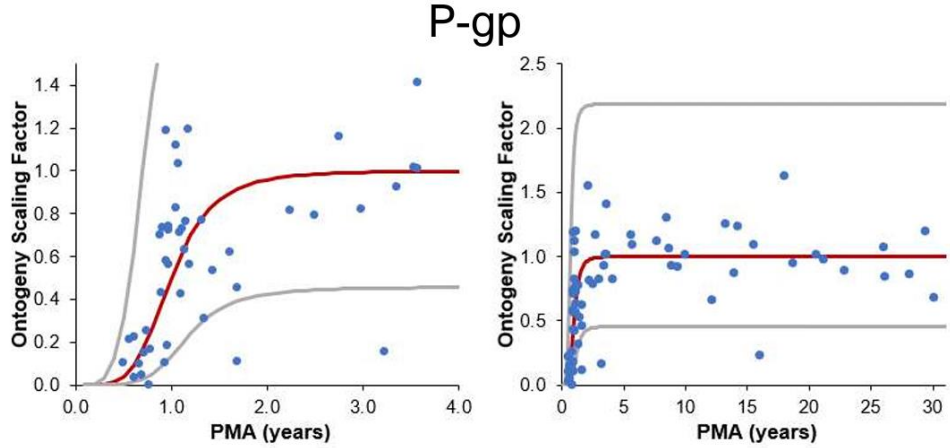


Figure adapted from Hunt et al. 2024, Figure S3

A.1.2.2 Description for PK-Sim®

Values for P-gp ontogeny are based on a published ontogeny (Hunt et al. 2024) which is based on transporter expression measurements (protein and mRNA) from human postmortem renal cortical tissue samples (Cheung et al. 2019).

A.1.2.3 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for P-gp (Geometric Mean)	Geometric Standard Deviation
0.1	0	2.903
0.2	0.001	2.901
0.3	0.004	2.89
0.4	0.015	2.857
0.5	0.04	2.783
0.6	0.088	2.653
0.7	0.163	2.472
0.8	0.264	2.267
0.9	0.381	2.072
1	0.499	1.91
1.1	0.606	1.789
1.2	0.696	1.702
1.3	0.767	1.64
1.4	0.822	1.597
1.5	0.864	1.567
1.6	0.895	1.546
1.7	0.918	1.53
1.8	0.936	1.519
1.9	0.949	1.511
2	0.959	1.504
2.1	0.967	1.5
2.2	0.973	1.496
2.3	0.978	1.493
2.4	0.982	1.491
2.5	0.985	1.489
2.6	0.987	1.487
2.7	0.989	1.486
2.8	0.991	1.485
2.9	0.992	1.484
3	0.993	1.484
4	0.998	1.481
5	0.999	1.48
10	1	1.48
15	1	1.48
20	1	1.48
30	1	1.48
40	1	1.48
60	1	1.48
80	1	1.48
100	1	1.48

4.8 Plasma Proteins

A.1 Human Serum albumin (HSA)

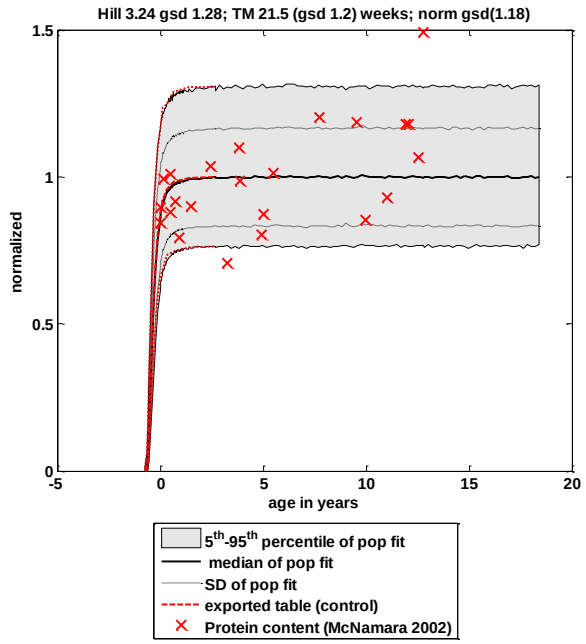
A.1.1 Original data

Reference	Used for fit
AAPS PharmSci. 2002;4(1):E4.; Protein binding predictions in infants.; McNamara PJ1, Alcorn J. Digital Object Identifyer (DOI): 10.1208/ps040104 → Figure 1 (serum protein concentrations)	 Yes

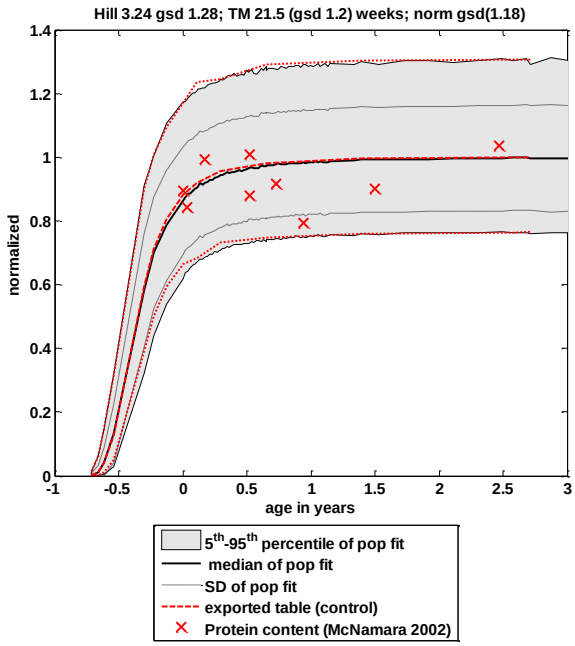
A.1.2 Fit results

A.1.2.1 Figure

Full range-plot



Zoom-in, x-axis



A.1.2.2 Fit function

$A = PMA^n/(A_{0.5}{}^n + PMA^n)$

With:
PMA = Post-menstrual age in weeks
A = Activity at PMA
$A_{0.5}$ = PMA at 50 % activity compared to adult
n = Hill coefficient
GeoSD = geometric standard deviation

Fit Result:
n = 3.240
$A_{0.5}$ = 21.533

GeoSD n = 1.284
GeoSD $A_{0.5}$ = 1.205
GeoSD adult = 1.177

GeoSD adult = geometric standard deviation in activity observed in adults

A.1.2.3 Description for PK-Sim®

Values for Human Serum albumin ontogeny are based on information to age dependency of serum albumin concentrations, as reported by McNamara et al., AAPS PharmSci. 2002;4(1):E4.

A.1.2.4 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for HSA	Standard Deviation
0.05	0.0011	4.46
0.1	0.011	2.8
0.16	0.04	2.2
0.23	0.13	1.74
0.46	0.59	1.29
0.55	0.71	1.24
0.64	0.81	1.2
0.77	0.88	1.19
0.88	0.92	1.2
1.06	0.96	1.17
1.42	0.98	1.18
2.17	1	1.18

A.2 Alpha-1-acid glycoprotein (AAG)

A new ontogeny for AAG has been implemented. This ontogeny is recently published at the PAGE 2018 Conference which is linked below:

Mayer et al., A novel approach to estimate ontogenies for PBPK applications – From literature data to simulations.

https://www.page-meeting.org/default.asp?abstract=8583

A.2.1.1 PK-Sim® Table

Post menstrual age [year(s)]	Ontogeny for AAG	Standard Deviation
0.766712298	0.254563711	1.829152038
0.770545289	0.288103739	1.881821245
0.77437828	0.320476631	1.840358216
0.785877253	0.404892648	1.715701843
0.797376226	0.47258055	1.628946923
0.805042209	0.510390851	1.588351585
0.816541182	0.558546687	1.54513096
0.831873146	0.610479462	1.508784277
0.847205111	0.652062602	1.486738469
0.889368013	0.732510638	1.459102868
0.946862879	0.798310922	1.447694197
1.035021675	0.855224822	1.443135263
1.169176363	0.900405864	1.441661523
1.387656856	0.935748949	1.441286708
1.721127082	0.96008023	1.441272349
3.032010039	0.987428363	1.441419782
13.60711841	1.002010167	1.441462514
30.76659822	1	1.440946066
90.45859001	0.986596986	1.443474188